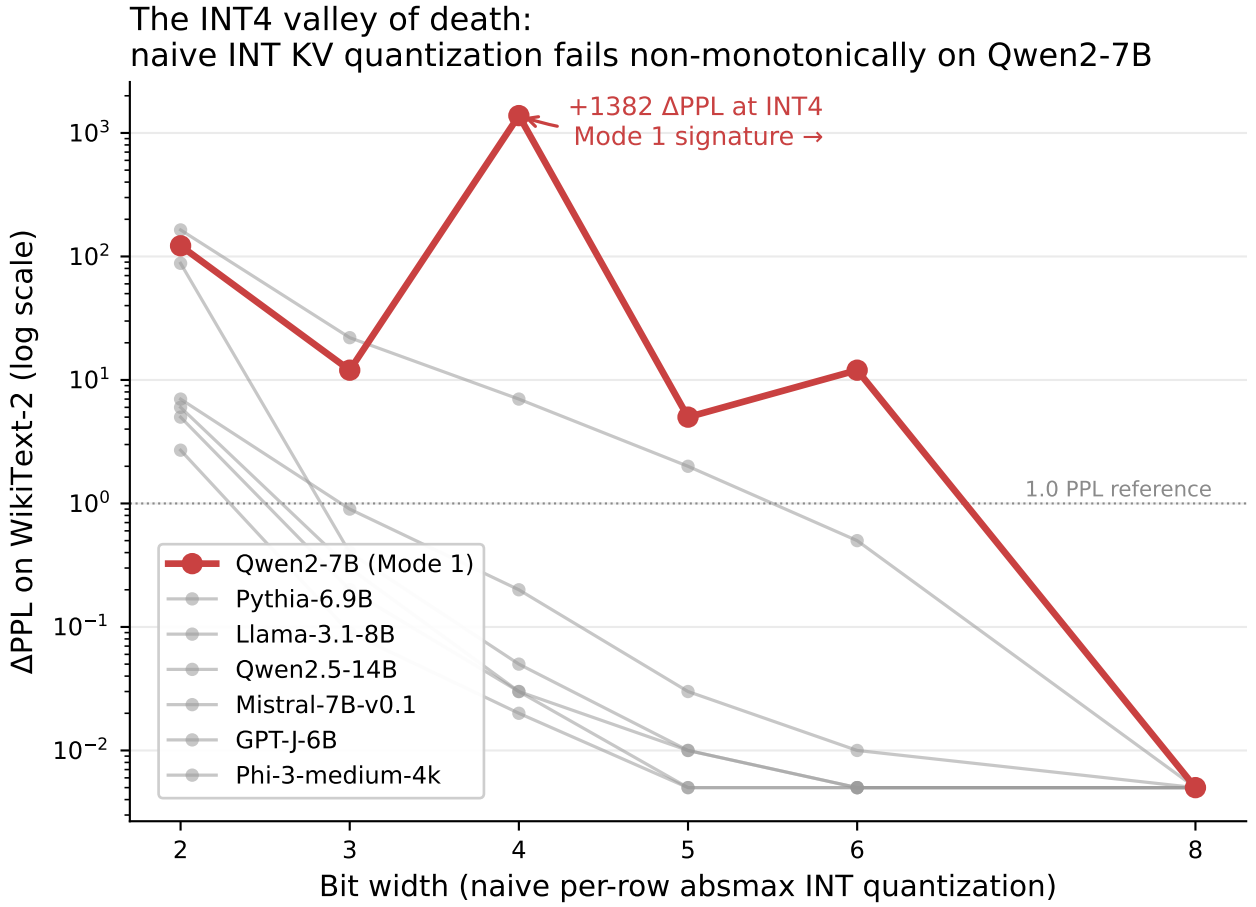
The INT4 valley of death:
naive INT KV quantization fails non-monotonically on Qwen2-7B
+1382 ΔPPL at INT4
Mode 1 signature →
1.0 PPL reference
Qwen2-7B (Mode 1)
Pythia-6.9B
Llama-3.1-8B
Qwen2.5-14B
Mistral-7B-v0.1
GPT-J-6B
Phi-3-medium-4k
ΔPPL on WikiText-2 (log scale)
Bit width (naive per-row absmax INT quantization)